
MODEL SELECTION AND VALIDATION REPORT (remove_7_biomass model)

OBSERVATIONS:1690

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_7_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.45	0.16
remove_7_tric	0.55	0.03
remove_7_teve	-0.41	0.03
sbt_mean	0.21	0.09
sbt_sd	0.06	0.05
sst_sd	0.04	0.04
log_chla_mean	0.06	0.04
depth	0.23	0.05
log_sum_fishing	0.07	0.03

Dispersion parameter: 0.96

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 206.42

Spatiotemporal marginal AR1 SD: 0.81

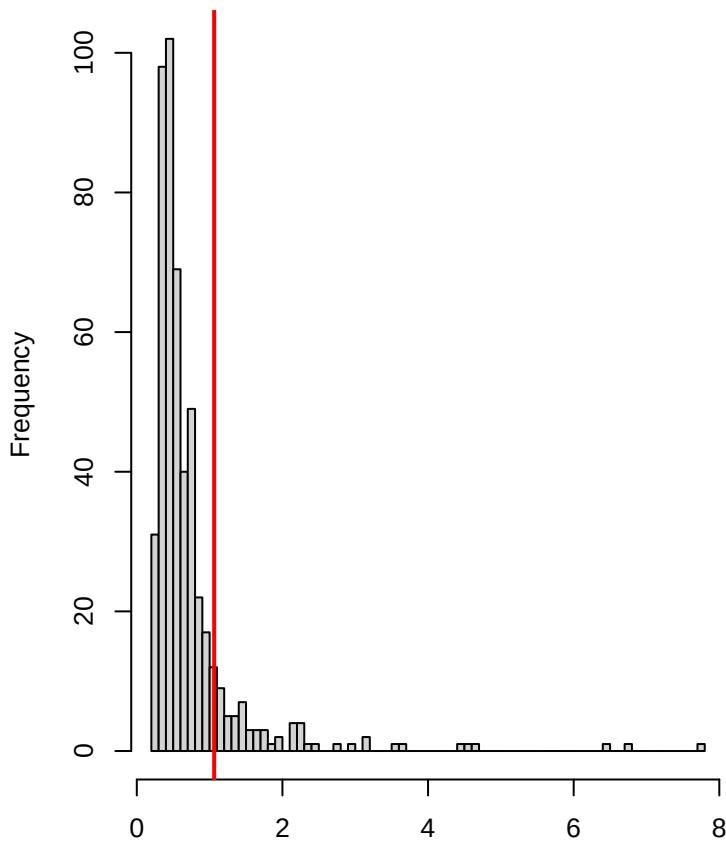
REML criterion at convergence: 10766.267

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

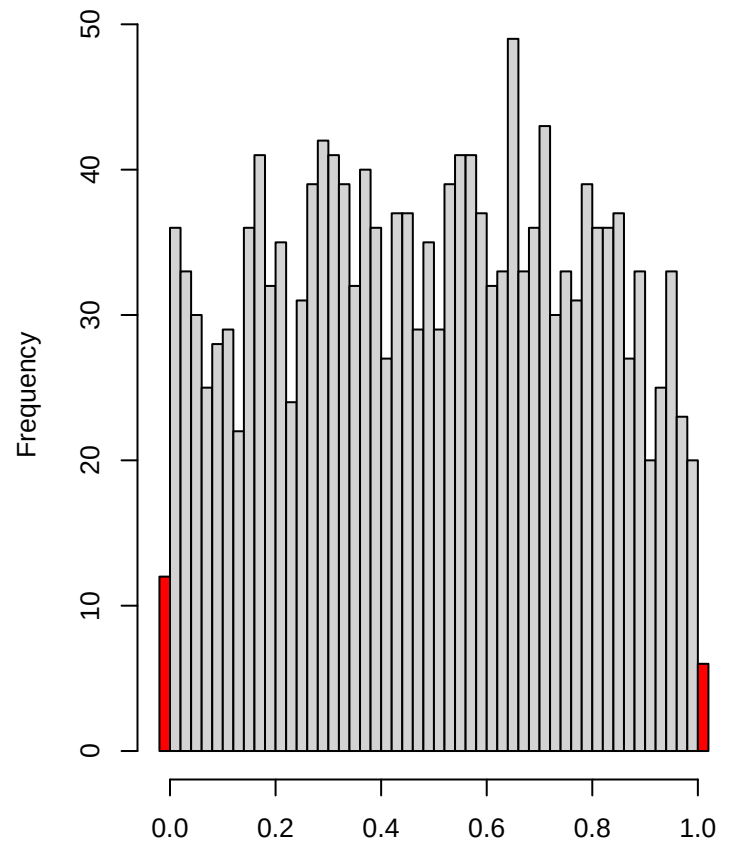
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	21558.53	2.655218
spatiotemporal_random_fields_ar1_with_anisotropy	15	21561.19	72.528672
spatial_random_fields	12	21633.72	3.784501
spatial_random_fields_with_anisotropy	14	21637.50	66.985439
spatiotemporal_random_fields_iid_with_anisotropy	14	21704.49	1.782388
spatiotemporal_random_fields_iid	12	21706.27	117.615262
error_only	10	21823.89	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



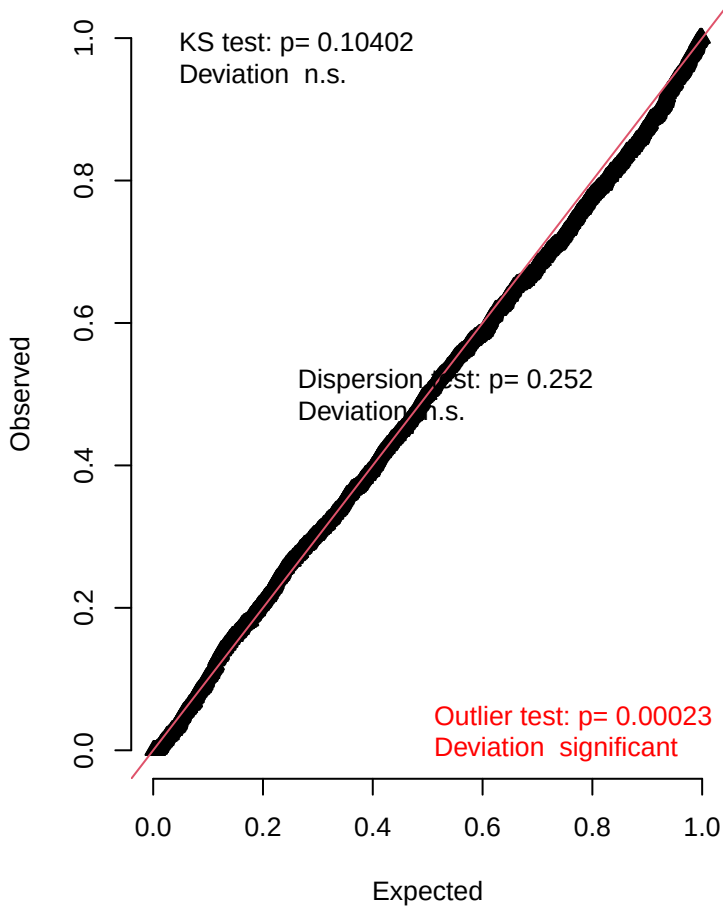
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

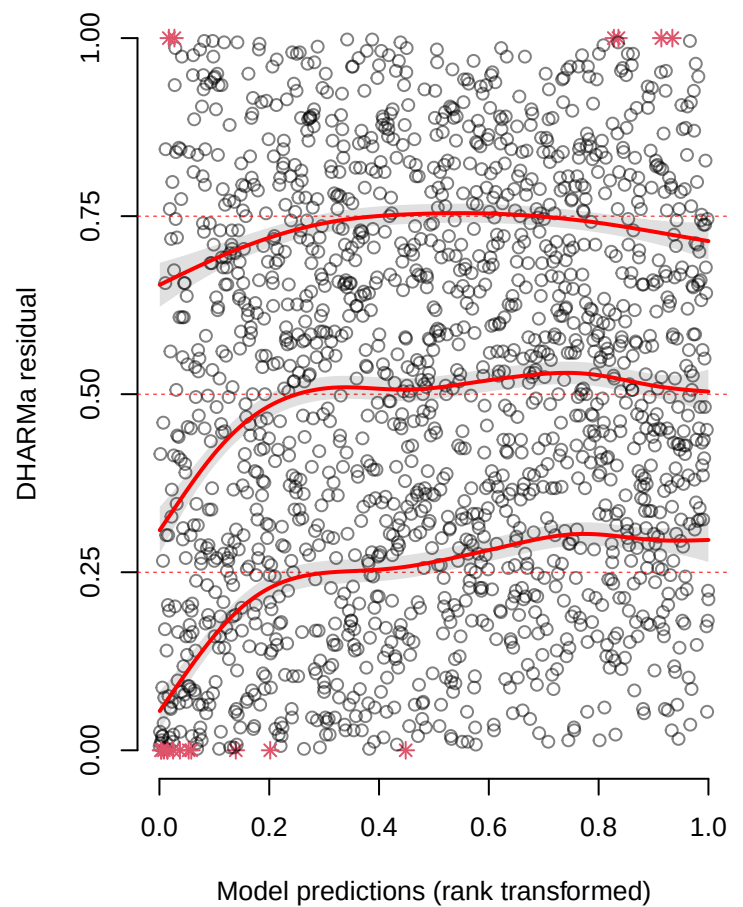


Residuals (outliers are marked red)

QQ plot residuals



DHARMA residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.014 | H1: Distance-based autocorrelation | p-value = 0.125

